

Practical-7 (MAD)

AIM :- To navigate between multiple activities and perform store-retrieve operation of simple data in Android using SharedPreferences.

- **SecondActivity.java :-**

```
package com.yourpackage.name; // Replace with your actual package name

import android.content.Context; import
android.content.SharedPreferences;
import android.os.Bundle; import
android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    private TextView textViewDisplay;

    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        textViewDisplay = findViewById(R.id.textViewDisplay);

        // Retrieve and display data as soon as the activity opens
        loadData();
    }

    private void loadData() {
        // Open the same SharedPreferences file using the constant from MainActivity
        SharedPreferences sharedPreferences =
        getSharedPreferences(MainActivity.SHARED_PREFS, Context.MODE_PRIVATE);

        // Get the string. The second parameter is a fallback default value if the key doesn't
        exist.
        String savedText = sharedPreferences.getString(MainActivity.TEXT_KEY, "No data
        found");

        // Display it in the TextView
        textViewDisplay.setText(savedText);
    }
}
```

- **MainActivity.java :-**

```
package com.yourpackage.name; // Replace with your actual package name
```

```
import android.content.Context; import
android.content.Intent; import
android.content.SharedPreferences; import
android.os.Bundle; import android.view.View;
import android.widget.Button; import
android.widget.EditText; import
android.widget.Toast; import
androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    private EditText editTextInput;
    private Button buttonSave, buttonNext;
```

```
    // Constants for SharedPreferences name and the key used to
    store data    public static final String SHARED_PREFS =
    "MySharedPrefs";    public static final String TEXT_KEY = "textKey";
```

```
    @Override    protected void onCreate(Bundle
    savedInstanceState)
    {        super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
```

```
        // Initialize UI components
    editTextInput = findViewById(R.id.editTextInput);
    buttonSave = findViewById(R.id.buttonSave);
    buttonNext = findViewById(R.id.buttonNext);
```

```
        // Save data when the save button is clicked
    buttonSave.setOnClickListener(new View.OnClickListener() {
        @Override            public
    void onClick(View v)
    {        saveData();
        }
    });
```

```
        // Navigate to the next activity
    buttonNext.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
    startActivity(intent);
        }
    });
}
```

```
    private void saveData() {
```

```

        // Initialize SharedPreferences in private mode
        SharedPreferences sharedPreferences = getSharedPreferences(SHARED_PREFS,
Context.MODE_PRIVATE);
        SharedPreferences.Editor editor = sharedPreferences.edit();

        // Put the string from the EditText into SharedPreferences
        editor.putString(TEXT_KEY, editTextInput.getText().toString());

        // apply() saves the data asynchronously (better performance than commit())
        editor.apply();

        Toast.makeText(this, "Data saved to SharedPreferences!",
Toast.LENGTH_SHORT).show();
    }
}

```

• AndroidManifest.xml :-

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">
    <application        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.MyApplication"> <activity
        android:name=".MainActivity"
        android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <activity
            android:name=".SecondActivity"
            android:exported="false" />

    </application>

</manifest>

```